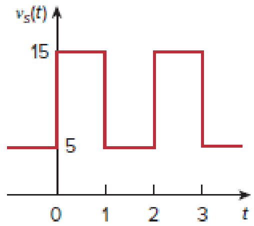
Chapter 17, Problem 39P

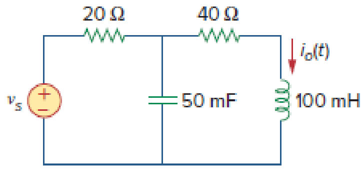
Problem

If the periodic voltage in Fig. 17.75(a) is applied to the circuit in Fig. 17.75(b), find *io*(*t*).

Figure 17.75:



(a)



(b)

Step-by-step solution

Step 1 of 9

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$\omega_0 = \dfrac{2\pi}{T}$ is the fundamental frequency.

The function is,

$$f(t) = \begin{cases} 15, & 0 < t < 1 \\ 5, & 1 < t < 2 \end{cases}$$

Step 2 of 9

The time period is, $T = 2$.

Determine the value of ω_0.

$$\omega_0 = \frac{2\pi}{2} \\ = \pi$$

Determine the value of a_0.

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt \\ = \frac{1}{2}\int_0^2 f(t)\,dt \\ = \frac{1}{2}\int_0^1 f(t)\,dt + \frac{1}{2}\int_1^2 f(t)\,dt$$

$$= \frac{1}{2}\int_0^1 15\,dt + \frac{1}{2}\int_1^2 5\,dt \\ = \frac{15}{2} + \frac{5}{2} \\ = 10$$

Step 3 of 9

Determine the value of a_n.

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_0 t)\,dt \\ = \frac{2}{2}\int_0^2 f(t)\cos(n\pi t)\,dt \\ = \int_0^1 15\cos(n\pi t)\,dt + \int_1^2 5\cos(n\pi t)\,dt$$

$$= \frac{15}{n\pi}\left[\sin(n\pi t)\right]_{t=0}^{t=1} + \frac{5}{n\pi}\left[\sin(n\pi t)\right]_{t=1}^{t=2} \\ = \frac{15}{n\pi}\sin(n\pi) + \frac{5}{n\pi}\left(\sin(2n\pi) - \sin(n\pi)\right) \\ = 0$$

Step 4 of 9

Determine the value of b_n.

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega_0 t)\,dt \\ = \frac{2}{2}\int_0^2 f(t)\sin(n\pi t)\,dt \\ = \int_0^1 15\sin(n\pi t)\,dt + \int_1^2 5\sin(n\pi t)\,dt$$

$$= \frac{15}{n\pi}\left[-\cos(n\pi t)\right]_{t=0}^{t=1} + \frac{5}{n\pi}\left[-\cos(n\pi t)\right]_{t=1}^{t=2} \\ = \frac{15}{n\pi}\left[1 - \cos(n\pi)\right] + \frac{5}{n\pi}\left[\cos(n\pi) - \cos(2n\pi)\right] \\ = \frac{15}{n\pi}\left[1 - \cos(n\pi)\right] + \frac{5}{n\pi}\left[\cos(n\pi) - 1\right] \\ = \frac{10}{n\pi}\left[1 - \cos(n\pi)\right]$$

The Fourier series is,

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right) \\ &= 10 + \sum_{n=1}^{\infty}\left(b_n \sin n\omega_0 t\right) \\ &= 10 + \sum_{n=1}^{\infty}\left\{\frac{10}{n\pi}\left[1-\cos(n\pi)\right]\sin(n\pi t)\right\} \end{aligned}$$

Refer to Figure 17.75 in the textbook.

Use Kirchhoff's voltage law in the frequency domain.

$$\begin{aligned} \mathbf{I}_s &= \frac{1}{20 + \dfrac{(40 + j\omega 0.1)\left(\dfrac{20}{j\omega}\right)}{40 + j\omega 0.1 + \dfrac{20}{j\omega}}}\mathbf{V}_s \\ &= \frac{1}{20 + \dfrac{20(40 + j\omega 0.1)}{j\omega 40 + 20 - 0.1\omega^2}}\mathbf{V}_s \\ &= \frac{j\omega 40 + 20 - 0.1\omega^2}{20\left(j\omega 40 + 20 - 0.1\omega^2\right) + 20(40 + j\omega 0.1)}\mathbf{V}_s \\ &= \frac{j\omega 40 + 20 - 0.1\omega^2}{j\omega 802 + 1200 - 2\omega^2}\mathbf{V}_s \end{aligned}$$

Comments (1)

Anonymous

please tell me this is not a serious chegg step

By current division, I_o can be expressed as follows:

$$\begin{aligned} \mathbf{I}_o &= \frac{\left(\dfrac{20}{j\omega}\right)}{40 + j\omega 0.1 + \dfrac{20}{j\omega}}\mathbf{I}_s \\ &= \frac{20}{j\omega 40 + 20 - 0.1\omega^2}\mathbf{I}_s \\ &= \frac{20}{j\omega 40 + 20 - 0.1\omega^2}\cdot\frac{j\omega 40 + 20 - 0.1\omega^2}{j\omega 802 + 1200 - 2\omega^2}\mathbf{V}_s \\ &= \frac{20}{j\omega 802 + 1200 - 2\omega^2}\mathbf{V}_s \end{aligned}$$

Substitute $\omega_n = n\pi$ in the equation.

$$\mathbf{I_o} = \frac{20}{jn\pi 802 + 1200 - 2n^2\pi^2} \times \mathbf{V}_s$$

For the n-th harmonic,

$$v_s(t) = 10 + \sum_{n=1}^{\infty}\left\{\frac{10}{n\pi}\left[1-\cos(n\pi)\right]\sin(n\pi t)\right\}$$

$$\mathbf{V_s} = \frac{10}{n\pi}\left[1-\cos(n\pi)\right]\angle -90°$$

For dc component, $\omega_n = 0 \text{ or } n = 0$.

$$\begin{aligned}\mathbf{I_{odc}} &= \lim_{n\to 0}\left|\frac{20}{jn\pi 802 + 1200 - 2n^2\pi^2} \times \mathrm{V_{sdc}}\right| \\ &= \lim_{n\to 0}\left|\frac{20}{1200} \times 10\right| \\ &= \frac{1}{6}\end{aligned}$$

Determine the expression for $\mathbf{I_o}$.

$$\begin{aligned}\mathbf{I_o} &= \frac{20}{j\omega 802 + 1200 - 2\omega^2}\mathbf{V_s} \\ &= \frac{20}{jn\pi 802 + 1200 - 2n^2\pi^2}\mathbf{V_s} \\ &= \frac{20}{jn\pi 802 + 1200 - 2n^2\pi^2} \cdot \frac{10}{n\pi}\left[1-\cos(n\pi)\right]\angle -90° \\ &= \frac{20}{\sqrt{(n\pi 802)^2 + (1200-2n^2\pi^2)^2}\angle \tan^{-1}\left(\frac{802n\pi}{1200-2n^2\pi^2}\right)} \cdot \frac{10}{n\pi}\left[1-\cos(n\pi)\right]\angle -90°\end{aligned}$$

$$= \frac{200 \times 2\angle\left[-90° - \tan^{-1}\left(\frac{802n\pi}{1200-2n^2\pi^2}\right)\right]}{n\pi\sqrt{(n\pi 802)^2 + (1200-2n^2\pi^2)^2}}, \; n \text{ is odd}$$

$$= \frac{400\angle\left[-90° - \tan^{-1}\left(\frac{802n\pi}{1200-2n^2\pi^2}\right)\right]}{n\pi\sqrt{(n\pi 802)^2 + (1200-2n^2\pi^2)^2}}, \; n \text{ is odd}$$

$$= \frac{400\angle\left[-90° + \tan^{-1}\left(\frac{802n\pi}{2n^2\pi^2 - 1200}\right)\right]}{n\pi\sqrt{(n\pi 802)^2 + (1200-2n^2\pi^2)^2}}, \; n \text{ is odd}$$

In the time domain, determine the expression for $\mathbf{I_o}$.

$$i_o(t)=\frac{1}{6}+\frac{400}{\pi}\sum_{k=1}^{\infty}\frac{\cos\left[n\pi t+\left[-90^\circ+\tan^{-1}\left(\frac{802n\pi}{2n^2\pi^2-1200}\right)\right]\right]}{n\sqrt{(n\pi 802)^2+(1200-2n^2\pi^2)^2}},n=2k-1$$

$$=\frac{1}{6}+\frac{400}{\pi}\sum_{k=1}^{\infty}\frac{\sin\left[n\pi t+\tan^{-1}\left(\frac{802n\pi}{2n^2\pi^2-1200}\right)\right]}{n\sqrt{(n\pi 802)^2+(1200-2n^2\pi^2)^2}},n=2k-1$$

$$=\frac{1}{6}+\frac{400}{\pi}\sum_{k=1}^{\infty}\frac{\sin\left[n\pi t+90^\circ-\cot^{-1}\left(\frac{802n\pi}{2n^2\pi^2-1200}\right)\right]}{n\sqrt{(n\pi 802)^2+(1200-2n^2\pi^2)^2}},n=2k-1$$

$$=\frac{1}{6}+\frac{400}{\pi}\sum_{k=1}^{\infty}\frac{\sin\left[n\pi t+90^\circ-\tan^{-1}\left(\frac{2n^2\pi^2-1200}{802n\pi}\right)\right]}{n\sqrt{(n\pi 802)^2+(1200-2n^2\pi^2)^2}},n=2k-1$$

Thus, the output current $i_o(t)$ is,

$$\boxed{i_o(t)=\frac{1}{6}+\frac{400}{\pi}\sum_{k=1}^{\infty}\frac{\sin\left[n\pi t+90^\circ-\tan^{-1}\left(\frac{2n^2\pi^2-1200}{802n\pi}\right)\right]}{n\sqrt{(n\pi 802)^2+(1200-2n^2\pi^2)^2}},n=2k-1}.$$

Comments (1)

Anonymous

problem needs to be updated for the new edition of the book.